MIL-M-12697A(MU)
17 March 1970
SUPERSEDING
MIL-M-12697(Ord)
13 May 1953

MILITARY SPECIFICATION

MOUNT, TELESCOPE, M21A1

1. SCOPE

1.1 This specification covers one type of telescope mount primarily designed to provide adjustments in elevation and to compensate for angles of cant for indirect fire.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Frankford Arsenal

F7578396	Mount, Telescope, M21A1

PACKAGING DATA SHEET

7578396	Packaging of Mount, Telescope, M21A1

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(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The mount shall be manufactured in accordance with Drawing F7578396 and all drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to and compliance with the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.3 Environmental.

3.3.1 Operating temperature.- The mount shall meet the applicable requirements of 3.9 while exposed to and thermally stabilized at ambient temperature of -40° and +150° Fahrenheit (F). Upon return to standard ambient temperature of +60° to +90°F, the mount shall meet the requirements of 3.3.2 to 3.9 inclusive.

3.3.2 Vibration.- The mount shall withstand a vibration frequency of 30 cycles per second (cps) ±5 at an amplitude of 1/16 inch (1/8 inch total excursion) for 5 minutes ± 15 seconds in each of two perpendicular planes. There shall be no evidence of damage or physical failure after vibration and the mount shall meet the following requirements.

3.4 Orientation.- The requirements of 3.5 to 3.9 inclusive shall be met with the mount oriented in the following position:

(a) The mounting surface of the bracket shall be horizontal within 0.3 mil.
(b) The axis of the telescope socket shall be vertical within 1.0 mil and the top surface of the socket horizontal within 0.3 mil.
(c) The elevation and cross level vial bubbles shall be centered within the width of a graduation line.
(d) The coarse and fine indices adjusted to coincide.
(e) A dummy load of 5 1/2 lbs. ± 1/8 lb. (to simulate weight of an M12A7 Series Panoramic Telescope) secured in the telescope socket.
(f) The adjusting screws in the telescope socket shall be set to bring the line of sight of a panoramic telescope parallel to a plumb line within 0.5 mil.

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3.5 Performance.

3.5.1 Excursion ranges.

3.5.1.1 Elevation travel.- Actuation of the elevation knob shall rotate the rocker arm through an excursion of at least 1100 mils in depression (elevation) and 178 mils in elevation (depression) from the position established in 3.4.

3.5.1.2 Cross level travel (cant).- Actuation of the cant correction knob shall move the telescope socket a minimum of 142 mils to the left and 142 mils to the right from the position established in 3.4.

3.6 Accuracy.

3.6.1 Plumb travel.- A line of sight shall track a plumb line within 0.5 mil when the mount is subjected to the requirements of 3.5.1.1.

3.6.2 Elevation level.- The elevation vial bubble shall remain central within one half a vial graduation when the mount is subjected to the requirements of 3.5.1.2.

3.6.3 Cross level.- The cross level vial bubble shall remain central within one half a vial graduation when the mount is subjected to the requirements of 3.5.1.1.

3.6.4 Backlash

3.6.4.1 Elevation.- Backlash in the elevation mechanism (3.5.1.1) shall not exceed 0.5 mil at 0° elevation.

3.6.4.2 Cross level (cant).- Backlash in the cross level (3.5.1.2) mechanism shall not exceed 0.5 mil at 0° elevation and cant.

3.7 Rigidity.

3.7.1 Elevation.- With the mount oriented in accordance with 3.4, gradually apply twenty five pounds pressure to the top surface, at the rear of the telescope socket, perpendicular to the vertical axis. Release the pressure and record movement. Apply the same pressure, in the same manner, to the front of the telescope socket. Release the pressure and record movement. The total movement shall not exceed 1.0 mil.

3.7.2 Cross level.- With the mount oriented in accordance with 3.4, gradually apply twenty five pounds pressure to the top surface, on the right hand side of the telescope socket, perpendicular to the vertical axis. Release the pressure and record movement. Apply the same pressure, in the same manner, to the left hand side of the telescope socket. Release the pressure and record movement. The total movement shall not exceed 1.0 mil.

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3.8 Azimuth correction.- The mount shall give left and right azimuth correction readings in accordance with the requirements and values of Table I.

TABLE I

AZIMUTH CORRECTION

Cant Angle (L&R) (In Mils)	Elevation (In mils)	Azimuth Correction (In mils)
124.4	500.0	66.3 ± 1.0
124.4	800.0	123.5 ± 1.4
124.4	1100.0	228.3 ± 2.1

3.9 Operability.- The operating torques shall be within the values specified in Table II at the temperatures indicated therein. The level vial covers and the wing knob (telescope socket) shall be operable at the temperatures specified in Table II. The vial covers shall completely cover and completely expose (detent to detent) their respective vials and the wing knob shall be operable from stop to stop.

TABLE II

OPERATING TORQUE VALUES

Control	(+60° to +90°F) and +150°F	-40°F
Elevation knob	4 lb in min. to 12 lb in max.	25 lb in max.
Cross level knob	4 lb in min. to 12 lb in max.	25 lb in max.

3.10 Workmanship.- The workmanship requirements of MIL-F-13926 shall be met. Exterior protective surface finishes shall be free of defects which may affect their protective qualities or the appearance of the mount. The mounting and locating surfaces shall be free of paint, nicks or burrs which may interfere with mounting accuracy.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) mounts and three (3) each of every component and subassembly covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicabel Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables III and IV herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

TABLE III - CLASSIFICATION OF DEFECTS

CRITICAL: NONE MAJOR: AQL 1.0% Defective	REQUIREMENT	TEST PROCEDURE
101. Vibration	3.3.2	4.6.2
102. Elevation travel	3.5.1.1	4.6.3.1
103. Crosslevel travel (cant)	3.5.1.2	4.6.3.2
104. Plumb travel	3.6.1	4.6.3.3
105. Elevation level	3.6.2	4.6.4
106. Crosslevel	3.6.3	4.6.4.1
107. Backlash - elevation	3.6.4.1	4.6.5
108. Backlash - crosslevel (cant)	3.6.4.2	4.6.5.1
109. Rigidity - elevation	3.7.1	4.6.6
110. Rigidity - crosslevel	3.7.2	4.6.6.1
111. Azimuth correction	3.8	4.6.7
112. Operability	3.9	4.6.8
113. Workmanship	3.10	4.6.9
MINOR: NONE		

NOTE: The tests of Table III shall be conducted at a temperature between +60° and +90°F.

4.4 Special sampling.- Sampling inspection for the characteristics listed in the following table shall be in accordance with inspection level S3, Table I of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5, Table II-A of MIL-STD-105. Samples selected shall be from units which have passed all other requirements of this specification.

TABLE IV

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Applicable dwgs - Visual
302. General specification	3.2	MIL-F-13926 - Visual
303. Operating temperature	3.3.1	4.6.1
304. Operability - elevation knob -40°F and +150°F	3.9	4.6.8
305. Operability - crosslevel knob -40°F and +150°F	3.9	4.6.8

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4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Operating temperatures.- The mounts shall be subjected to the temperatures specified in 3.3.1 utilizing the test facilities referenced in 4.5.2.2. The temperature shall be changed gradually to avoid thermal shock. When the mount has been thermally stabilized at each extreme temperature, it shall meet the requirements of 3.9. Upon return to standard ambient temperature, the mount shall be tested in accordance with all remaining tests specified herein to determine compliance with 3.3.1.

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4.6.2 Vibration.- The mount shall be mounted on a vibration machine equipped with facilities to securely hold the mount while the machine is in motion. The mount shall be vibrated in accordance with 3.3.2 at the frequency, amplitude and duration specified. Subsequent to vibration, a visual examination shall be made to determine evidence of physical failure or damage and the mount shall then meet the requirements specified in 3.3.2.

4.6.3 Orientation.- A final acceptance test fixture with a suitable adapter shall be utilized for the subsequent tests specified herein. The fixture and the adapter shall conform to the accuracies specified on Drawings D6253553 and D6510877. A wall target shall be established in front of the fixture. The target shall contain calibrated graduations to directly measure any angular deviations in elevation, crosslevel (cant) and plumb travel. A collimating telescope shall be attached to the test fixture so that the longitudinal axis of the telescope is 90° to the axis about which the fixture rotates in elevation. Rotate the fixture until the telescope's line of sight coincides with a plumb line. Set the indices of the crossleveling and elevation scales to indicate zero. Remove the collimator.

a. Attach the adapter to the test fixture and with a precision level, ascertain that the adapter's mounting surface is level in both planes. Position the mount on the adapter as specified in 3.4. Attach the auxiliary telescope on the flat surface of the mount's telescope socket. The position of the auxiliary telescope's optical axis shall simulate the optical axis of a panoramic telescope M12A7, Drawing F7687260. Adjust the telescope's optical axis to coincide with the zero elevation and crosslevel graduations on the wall target.

b. With the telescope's optical axis established and the fixture's elevation and crosslevel indices in zero position and the mount oriented as specified, the following tests shall be performed.

4.6.3.1 Elevation travel.- Rotate the test fixture to depress a minimum of 1100 mils. Actuate the mount's elevation knob in a reverse direction to its full travel. The optical axis shall return, at a minimum, to the established zero elevation position.

a. Reset the optical axis to zero elevation. Reset the fixture's elevation and crosslevel indices to zero. Rotate the fixture to elevate a minimum of 178 mils. Counter rotate the mount's elevation knob to its full travel. The optical axis shall return, at a minimum, to the established zero elevation.

4.6.3.2 Crosslevel travel (cant).- Reset the fixture and the optical axis as specified above. Rotate the fixture a minimum of 142 mils in the right direction. Counter rotate the mount's cant correction knob to its full travel. The optical axis shall return, at a minimum, to the established zero position.

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a. Reset the fixture and the optical axis to the zero elevation and crosslevel position. Rotate the fixture a minimum of 142 mils in the left direction. Counter rotate the mount's cant correction knob to its full travel. The optical axis shall return, at a minimum, to the established zero position.

4.6.3.3 Plumb travel.- Reset the optical axis and the fixture to the zero positions. Depress the fixture 200 mils. Counter rotate the mount's elevation knob to bring the line of sight back to the zero elevation on the target. Observe the position of the vertical line of sight while tracking the target plumb line. Repeat this procedure, at 200 mil increment settings. The line of sight shall track the plumb line throughout the full 1100 mil elevation excursion within the limits specified in 3.6.1.

a. Reset the optical axis and the fixture to the zero positions. Elevate the fixture 178 mils. Counter rotate the mount's elevation knob to bring the line of sight back to the zero elevation. Observe the position of the vertical line of sight while tracking the plumb line. The line of sight shall track the plumb line throughout the full 178 mil depression within the limits specified in 3.6.1.

4.6.4 Elevation level.- During the performance of the test specified in 4.6.3.2, the position of the elevation vial bubble shall be observed. The vial bubble shall remain central within the limits specified in 3.6.2.

4.6.4.1 Crosslevel.- During the performance of the test specified in 4.6.3.1, the position of the crosslevel vial bubble shall be observed. The vial bubble shall remain central within the limits specified in 3.6.3.

4.6.5 Backlash - elevation.- Reset the test fixture and telescope's optical axis to zero elevation and crosslevel. The elevation level vial bubble shall be centered. Rotate the elevation fixture 100 mils in depression. Counter rotate the mount's elevation knob to bring the elevation level vial bubble to the position previously established. Record the position of the optical axis with respect to the zero elevation graduation on the target. The difference in the recorded position of the optical axis with respect to the optical axis established at zero elevation constitutes the measured backlash. The backlash shall not exceed the limits specified in 3.6.4.1. Repeat this test at two additional 100 mil increments in elevation and two 50 mil increments in depression of the mount.

4.6.5.1 Backlash - crosslevel cant.- Reset the test fixture and the telescope's optical axis to zero elevation and crosslevel position. The crosslevel vial bubble shall be centered. Scribe a line from the worm housing onto the periphery of the crossleveling knob. This will result in two coinciding lines to be used as indices. Rotate the crossleveling knob to position the telescope's optical axis in coincidence with the 142 mil right graduation on the target. Rotate the crossleveling knob in a reverse direction until the scribed index lines coincide. Record the position of

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of the optical axis with respect to the zero graduation. The difference in the recorded position of the optical axis with respect to the optical axis established at zero elevation and crosslevel position constitutes the measured backlash. The backlash shall not exceed the limits specified in 3.6.4.2. Repeat this test in rotating the crosslevel knob to the 142 mil left graduation on the target.

4.6.6 Rigidity - elevation.- Reset the test fixture and the telescope's optical axis to zero elevation and crosslevel position. Using a force gage, apply a pressure specified in 3.7.1. Record any displacement of the optical axis with respect to the target's zero elevation position. The total displacement in both the front and rear directions shall be within the limits specified in 3.7.1.

4.6.6.1 Rigidity - crosslevel.- Reset the test fixture and the telescope's optical axis as specified above. Apply the specified pressure to the top surface in both right and left directions. Record any displacement of the optical axis with respect to the target. The total displacement shall be within the limits specified in 3.7.2.

4.6.7 Azimuth correction.- Reset the test fixture and the telescope's optical axis to zero elevation and crosslevel position. The crosslevel and elevation level vial bubbles shall be centered. Rotate the test fixture to elevate 500 mils. Rotate the fixture's cant mechanism to read 124.4 mils in the left direction. Counter rotate the cant correction knob of the mount and center the crosslevel bubble. Check the elevation vial bubble and read the azimuth correction on the calibrated wall target. The azimuth correction shall be within the limits specified in Table I. Repeat this test at 800 mils and 1100 mils in elevation with the mount canted 124.4 mils to the left and 124.4 mils to the right of the zero elevation position. All azimuth corrections in the three elevation positions shall be within the limits specified in Table I.

4.6.8 Operability.- This test shall be performed by visual and tactile means at the temperatures specified in 3.9 and Table II. A torque wrench and a suitable adapter shall be utilized to apply the specified torque to the elevation and crosslevel knobs. The level vial covers and wing knob of the telescope socket shall be operated by normal hand pressure. The movements shall be smooth and uniform without binding.

4.6.9 Workmanship.- This test shall be performed by visual means for compliance with the requirements specified in 3.10.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 7578396. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use.- The Telescope Mount M21A1 is primarily used for indirect fire on the 105MM (Towed) Howitzer M101A1. The mount compensates for deflection errors which are introduced when the weapon is elevated with the trunnion canted.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification
b. Selection of applicable levels of preservation, packaging and packing
c. Applicable stock number

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